

Regynix

Cadence Genus Retiming Synthesis Flow

Automated Register Retiming for Timing Closure

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Cadence Genus 21.14

RTL-to-Gates Flow

What This Project Does

OVERVIEW

Automated RTL Synthesis with Register Retiming

01

Read RTL Verilog

Ingests testdesign.v – a multi-output FIR-like datapath with 4 parallel processing streams

02

Apply Timing Constraints

Reads testdesign.sdc defining a 200 MHz clock (5 ns period), I/O delays, and false paths

03

Enable Retiming

Activates Genus register retiming to rebalance pipeline stages and reduce critical-path delay

04

Synthesize & Report

Runs syn_gen → syn_map → syn_opt and emits QoR, timing, area, and gate reports

Main Objective: Improve timing performance through automatic register retiming during RTL synthesis

Project Inputs

FILE STRUCTURE

Key Input Files & Their Purpose

File	Type	Purpose
testdesign.v	RTL Verilog	Top-level design — 4-input datapath with 16 FIR coefficients
testdesign.sdc	Timing Constraints	200 MHz clock, I/O delays (2.25 ns), setup uncertainty (1.25 ns)
LIB/*.lib	Standard Cell Library	5 timing libs: slow.lib, pll.lib, CDK_S128x16, CDK_S256x16, CDK_R512x16
run.tcl	Synthesis Script	Complete end-to-end Genus flow: read → elaborate → retime → syn_gen/map/opt

SDC Key Parameters

- Clock Period: 5.00 ns (200 MHz)
- Setup Uncertainty: 1.25 ns
- Input / Output Delay: 2.25 ns

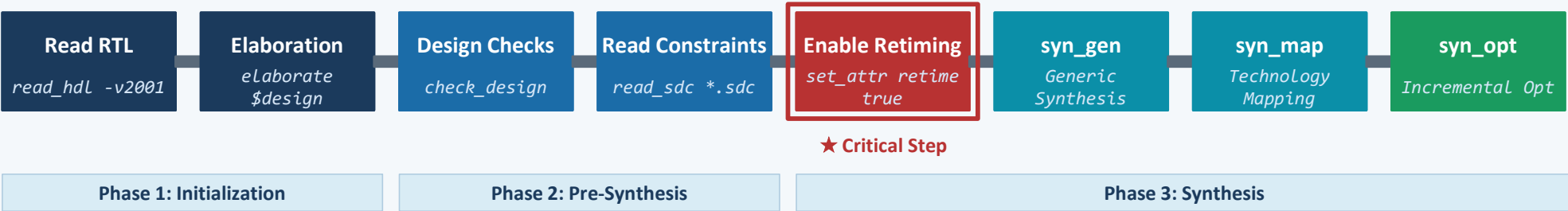
Design Statistics

- Module: testdesign (Verilog-2001)
- Outputs: 4 × 10-bit vectors (xout_15_1..4)
- Inputs: 4 × 5-bit data + 16 × 8-bit coefficients

Overall Synthesis Flow

ARCHITECTURE

Eight-Stage RTL-to-Gates Pipeline



Generated Outputs

- testdesign_generic.db
- testdesign_map.db
- testdesign_incr.db
- reports/ (QoR, timing, area, gates)

Retiming Effect

WNS Before
-4221 ps
WNS After (syn_opt)
-883 ps
TNS Improvement
-890k → -295k ps

run.tcl — Library Setup & Design Initialization

STEP 1-2

Configuring the Synthesis Environment

run.tcl – Library & Initialization

```
# Library setup
set_attribute library {
  ./LIB/slow.lib
  ./LIB/p11.lib
  ./LIB/CDK_S128x16.lib
  ./LIB/CDK_S256x16.lib
  ./LIB/CDK_R512x16.lib
} /

# Top-level module
set design testdesign

# Synthesis effort levels
set SYN_EFF medium
```

Standard Cell Libraries

5 libs loaded: timing-accurate slow-corner models plus SRAM/RAM macros (CDK_S128x16, CDK_S256x16, CDK_R512x16)

Top Module Declaration

set design testdesign — tells Genus which RTL module is the synthesis root

Effort Levels

'medium' effort balances runtime vs quality; can be raised to 'high' for timing closure

TNS Optimization

set_attribute tns_opto true — enables Total Negative Slack optimization across all violating paths

Debug flags: retime_debug and retime_debug_path are enabled → Genus writes retiming move logs to st_debug_logs/ for post-run analysis.

run.tcl — Read RTL, Elaboration & Design Checks

STEP 3-5

Parsing, Building the Hierarchy, and Validating the Design

RTL Read & Elaborate

```
# Read Verilog RTL
read_hdl -v2001 testdesign.v

# Build internal hierarchy
elaborate $design

# Check design integrity
check_design
check_design -undriven -unloaded \
  -multidriven -unresolved \
  -constant > check_design.report
```

*report timing -endpoints -num_paths 10 > CritPath_SDC.rpt ← pre-retiming
critical path baseline (WNS = -4221 ps)*

read_hdl -v2001

- Parses Verilog-2001 syntax
- Registers all modules and parameters
- Flags syntax errors immediately

elaborate \$design

- Resolves module hierarchy
- Expands parameters and generates nets
- Creates the internal netlist object

check_design + timing lint

- Detects undriven / unloaded signals
- Flags multidriven nets
- Reports constraint issues before synthesis

run.tcl — Reading SDC Timing Constraints

STEP 6

Defining Clock, I/O Delays, and False Paths

```
testdesign.sdc (excerpt)

set CLK_200_PERIOD      5.00
set CLK_200_S_UNCERT    1.25
set CLK_200_INPUT_DELAY 2.25
set CLK_200_OUTPUT_DELAY 2.25

create_clock -period 5.00 [get_ports clk]
set_clock_uncertainty -setup 1.25 clk

# Async reset – excluded from timing
set_false_path -from [get_ports rst_1]

# Synchronous inputs (xin_1..4)
set_input_delay 2.25 [get_ports $clk_inputs] \
    -clock [get_clocks clk]

# Outputs (xout_15 1..4)
```

read_sdc Constraints/testdesign.sdc ← TCL command loads the above constraints into the Genus timing engine

Constraint	Value	Impact
Clock Period	5.00 ns	Target frequency: 200 MHz
Setup Uncert.	1.25 ns	Guard-band reduces usable window to 3.75 ns
Input Delay	2.25 ns	Time consumed by upstream logic
Output Delay	2.25 ns	Time reserved for downstream logic
False Path	rst_1 + 16 coef ports	Excluded from timing analysis

⚠ Why Constraints Matter
Without SDC, Genus has no timing target and retiming has no objective — the tool cannot determine which register moves improve slack.

run.tcl — Enabling Register Retiming ★ Most Critical Step

STEP 7

Allowing Genus to Move Registers Across Combinational Logic

Retiming Activation Commands

```
# Enable retiming on all instances
set_attr retime true /des*/

# Allow reset logic to be retimed
set_attr retime_optimize_reset true /
```

BEFORE



-4221 ps

Imbalanced paths



AFTER



-883 ps

Balanced paths

Move Registers Forward

Pushes flip-flops past combinational gates to reduce logic depth on long paths

Move Registers Backward

Pulls flip-flops through gates to absorb slack from short, fast paths

Balance Pipeline Stages

Equalises delay across parallel paths so no single stage dominates

Reduce Critical Path

WNS improved from -4221 ps → -883 ps (79% reduction in worst violation)

run.tcl — Synthesis Stages: syn_gen → syn_map → syn_opt

STEP 8–10

Three Progressive Stages of RTL-to-Gates Synthesis

syn_gen

Generic Synthesis

Converts RTL to technology-independent Boolean logic. Applies initial retiming transformations — register moves determined here.

QoR Snapshot

- 23,402 leaf instances
- 72,981 area units
- WNS: -2688 ps / TNS: -890k ps

write_db → testdesign_generic.db

syn_map

Technology Mapping

Maps generic gates to actual standard cells (NAND, FF, buffers) using timing-accurate library data. Area drops ~50%.

QoR Snapshot

- 16,167 leaf instances
- 36,839 area units
- WNS: -2182 ps / TNS: -832k ps

write_db → testdesign_map.db

syn_opt

Incremental Optimisation

Final pass: buffer insertion, cell resizing, additional retiming. Largest improvement in WNS — slack cut by ~60% from map stage.

QoR Snapshot

- 22,799 leaf instances
- 44,881 area units
- WNS: -883 ps / TNS: -295k ps

write_db → testdesign_incr.db

WNS: -2688 ps

→

WNS: -2182 ps

→ WNS: -883 ps ✓ Best

Output Reports & Final Results

RESULTS

Understanding the Generated Reports

Report File	Stage	Key Information
gen_qor.rpt	syn_gen	WNS -2688 ps, TNS -890k, 23,402 cells, area 72,981
map_qor.rpt	syn_map	WNS -2182 ps, TNS -832k, 16,167 cells, area 36,839
iopt_qor.rpt	syn_opt	WNS -883 ps, TNS -295k, 22,799 cells, area 44,881
TimingPath_Opt.rpt	syn_opt	Top 10 critical endpoints; worst: csa_tree_add..._reg/D
CritPath_SDC.rpt	Pre-syn	Baseline WNS -4221 ps (before retiming)
map_area.rpt	syn_map	Detailed cell area and physical hierarchy
map_gates.rpt	syn_map	Gate-level statistics (NAND, FF, buffer counts)

Key Timing Metrics

Slack:

Required Time – Arrival Time. Positive = timing met.

WNS:

Worst Negative Slack — the most critical single path violation.

TNS:

Total Negative Slack — sum of all violating path slacks.

Conclusion

Regynix demonstrates how Cadence Genus performs automated retiming optimization — delivering a 79% improvement in worst-case timing slack through intelligent register redistribution across pipeline logic.

Key Concepts Demonstrated

RTL Synthesis Flow

Read → elaborate → check → constrain → synthesize

Retiming Mechanism

Genus moves FFs forward/backward through logic to balance pipeline

SDC Constraints

Clock (200 MHz), I/O delays, false paths define the timing target

Results Achieved

Technology Mapping

Generic Boolean → actual NAND/FF/buffer standard cells

QoR Analysis

WNS/TNS metrics across three synthesis stages show steady improvement

Result

WNS improved from -4221 ps (baseline) to -883 ps (syn_opt)